

SKILL-4

STORED PROCEDURES

1. PROCEDURE TO DISPLAY ALL CUSTOMERS

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
99 DELIMITER //
100 CREATE PROCEDURE GetAllCustomers()
101 BEGIN
102     SELECT * FROM Customer;
103 END //
104 DELIMITER ;
105 CALL GetAllCustomers();
106
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result 5 x

#	Time	Action	Message	Duration / Fetch
17	20:10:33	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
18	20:10:33	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
19	20:17:15	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * F...	0 row(s) affected	0.015 sec
20	20:17:15	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec

2. PROCEDURE TO DISPLAY ACCOUNT DETAILS FOR A GIVEN ACCOUNT NUMBER

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
107 DELIMITER //
108 CREATE PROCEDURE GetAccountDetails(
109     IN p_account_No INT
110 )
111 BEGIN
112     SELECT *
113     FROM Account
114     WHERE Account_No = p_Account_No;
115 END //
116 DELIMITER ;
117 CALL GetAccountDetails(10001);
118
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

Result 6 x

#	Time	Action	Message	Duration / Fetch
19	20:17:15	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * F...	0 row(s) affected	0.015 sec
20	20:17:15	CALL GetAllCustomers()	5 row(s) returned	0.016 sec / 0.000 sec
21	20:19:03	CREATE PROCEDURE GetAccountDetails(IN p_Account_N...	0 row(s) affected	0.031 sec

3. PROCEDURE TO FIND CUSTOMER ACCOUNTS

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

119 DELIMITER //
120 CREATE PROCEDURE GetCustomerAccounts(
121     IN p_Customer_ID INT
122 )
123 BEGIN
124     SELECT
125         C.Customer_ID,
126         C.Customer_Name,
127         A.Account_No,
128         A.Account_Type,

```

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

Result 7 x Read Only Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
26	20:23:08	CREATE PROCEDURE GetCustomerAccounts(IN p_Custom...	0 row(s) affected	0.016 sec
27	20:23:08	CALL GetCustomerAccounts(101)	1 row(s) returned	0.000 sec / 0.000 sec

4. PROCEDURE TO DEPOSIT MONEY

```

139 DELIMITER //
140 CREATE PROCEDURE DepositMoney(
141     IN p_Account_No INT,
142     IN p_Amount DECIMAL(12,2)
143 )
144 BEGIN
145     UPDATE Account
146     SET Balance = Balance + p_Amount
147     WHERE Account_No = p_Account_No;
148 END //
149 DELIMITER ;
150 CALL DepositMoney(10001, 5000);
151

```

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
28	20:24:46	CREATE PROCEDURE DepositMoney(IN p_Account_No IN...	0 row(s) affected	0.015 sec
29	20:24:46	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.016 sec

5. PROCEDURE TO WITHDRAW MONEY

```

152 DELIMITER //
153 CREATE PROCEDURE WithdrawMoney(
154     IN p_Account_No INT,
155     IN p_Amount DECIMAL(12,2)
156 )
157 BEGIN
158     UPDATE Account
159     SET Balance = Balance - p_Amount
160     WHERE Account_No = p_Account_No;
161 END //
162 DELIMITER ;
163 CALL WithdrawMoney(10001, 3000);
164

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
30	20:27:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No I...	0 row(s) affected	0.016 sec
31	20:27:26	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.000 sec

6. TRIGGER – PREVENT NEGATIVE BALANCE

```

166 DELIMITER //
167 CREATE TRIGGER CheckBalance
168 BEFORE UPDATE ON Account
169 FOR EACH ROW
170 BEGIN
171     IF NEW.Balance < 0 THEN
172         SIGNAL SQLSTATE '45000'
173         SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
174     END IF;
175 END //
176 DELIMITER ;
177 UPDATE Account
178 SET Balance = Balance - 1000
179 WHERE Account_No = 10001;
180

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	20:31:24	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Acco...	0 row(s) affected	0.032 sec
37	20:31:24	UPDATE Account SET Balance = Balance - 1000 WHERE Acc...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

7. CREATE TRANSACTION AUDIT TABLE

```

198 CREATE TABLE Transaction_Audit (
199     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
200     Transaction_ID INT,
201     Account_No INT,
202     Transaction_Type VARCHAR(20),
203     Amount DECIMAL(12,2),
204     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
205 );
206
207

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
43	21:07:33	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT...	Error Code: 1359. Trigger already exists	0.000 sec
44	21:08:23	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY ...	0 row(s) affected	0.062 sec

8. TRIGGER – TRANSACTION AUDIT

```

198 CREATE TABLE Transaction_Audit (
199     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
200     Transaction_ID INT,
201     Account_No INT,
202     Transaction_Type VARCHAR(20),
203     Amount DECIMAL(12,2),
204     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
205 );
206
207 DROP TRIGGER IF EXISTS TransactionAudit;
208
209 DELIMITER //
210

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
50	21:15:00	INSERT INTO Bank_Transaction (Account_No, Transaction_Ty...	1 row(s) affected	0.016 sec

9. TRIGGER – TRANSACTION AUDIT

disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

198 CREATE TABLE Transaction_Audit (
199     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
200     Transaction_ID INT,
201     Account_No INT,
202     Transaction_Type VARCHAR(20),
203     Amount DECIMAL(12,2),
204     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
205 );
206

```

	Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
▶	1	6	10001	DEPOSIT	2500.00	2026-08-13 21:15:00
•	NULL	NULL	NULL	NULL	NULL	NULL

Transaction_Audit 8 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 50	21:15:00	INSERT INTO Bank_Transaction (Account_No, Transaction_Ty...	1 row(s) affected	0.016 sec
✓ 51	21:15:00	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

10. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

241 DELIMITER //
242 CREATE TRIGGER UpdateBalanceAfterTransaction
243 AFTER INSERT ON Bank_Transaction
244 FOR EACH ROW
245 BEGIN
246
247     IF NEW.Transaction_Type = 'DEPOSIT' THEN
248
249         UPDATE Account
250         SET Balance = Balance + NEW.Amount
251         WHERE Account_No = NEW.Account_No;

```

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	56000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

Account 9 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 53	21:25:02	INSERT INTO Bank_Transaction (Account_No, Transaction_Ty...	1 row(s) affected	0.000 sec
✓ 54	21:25:02	SELECT * FROM Account WHERE Account_No = 10001 LIMIT ...	1 row(s) returned	0.000 sec / 0.000 sec

11. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE

```

257
258     UPDATE Account
259     SET Balance = Balance - NEW.Amount
260     WHERE Account_No = NEW.Account_No;
261
262 END IF;
263
264 END //
265
266 DELIMITER ;
267

```

disabled. Use the toolbar
manually get help for the
current caret position or to
toggle automatic help.

	Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
▶	1	6	10001	DEPOSIT	2500.00	2026-08-13 21:15:00
	2	7	10001	DEPOSIT	5000.00	2026-08-13 21:25:02
	3	8	10001	DEPOSIT	2500.00	2026-08-13 21:35:32
•	NULL	NULL	NULL	NULL	NULL	NULL

Transaction_Audit 10 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
63	21:35:32	DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction	0 row(s) affected	0.000 sec
64	21:35:32	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER IN...	0 row(s) affected	0.016 sec

12. PROCEDURE – ACCOUNT TRANSFER

```

268 DELIMITER //
269
270 CREATE PROCEDURE TransferMoney(
271     IN SenderAccount INT,
272     IN ReceiverAccount INT,
273     IN TransferAmount DECIMAL(12,2)
274 )
275 BEGIN
276
277     DECLARE SenderBalance DECIMAL(12,2);
278
279     SELECT Balance
280

```

Automatic context help is
disabled. Use the toolbar
manually get help for the
current caret position or to
toggle automatic help.

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	53500.00	Hyderabad
	10002	102	Savings	80000.00	Vijayawada
•	NULL	NULL	NULL	NULL	NULL

Account 11 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
66	21:37:58	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected	0.000 sec
67	21:37:58	SELECT * FROM Account WHERE Account_No IN (10001,100...	2 row(s) returned	0.000 sec / 0.000 sec

13. PROCEDURE – CUSTOMER LOAN DETAILS

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

309 DELIMITER //
310
311 CREATE PROCEDURE GetCustomerLoans(
312     IN p_Customer_ID INT
313 )
314 BEGIN
315
316     SELECT
317         C.Customer_Name,
318         L.Loan_ID,
319         L.Loan_Type

```

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Ravi Kumar	501	Home Loan	5000000.00	7.50

Result 12 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
68	21:40:18	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT)	0 row(s) affected	0.016 sec
69	21:40:18	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec

14. PROCEDURE – HIGH BALANCE ACCOUNTS

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

332 DELIMITER //
333
334 CREATE PROCEDURE HighBalanceAccounts(
335     IN MinimumBalance DECIMAL(12,2)
336 )
337 BEGIN
338
339     SELECT *
340     FROM Account
341     WHERE Balance >= MinimumBalance
342     ORDER BY Balance DESC

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10001	101	Savings	53500.00	Hyderabad

Result 13 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
70	21:42:14	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2))	0 row(s) affected	0.031 sec
71	21:42:14	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec

15. PROCEDURE WITH IN, OUT PARAMETERS

348 DELIMITER //

349

350 CREATE PROCEDURE GetBalance(
 351 IN p_Account_No INT,
 352 OUT p_Balance DECIMAL(12,2)
 353)
 354 BEGIN
 355
 356 SELECT Balance
 357 INTO p_Balance

Result Grid

@CurrentBalance
\$3500.00

Result 14 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
73	21:44:18	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.000 sec
74	21:44:18	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

16. TRIGGER MANAGEMENT COMMANDS

367 SHOW TRIGGERS;
 368 SHOW CREATE TRIGGER CheckBalance;
 369 DROP TRIGGER CheckBalance;
 370
 371
 372
 373
 374
 375
 376

Result Grid

Trigger	sql_mode	SQL Original Statement
CheckBalance	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	CREATE DEFINER='root'@'localhost' TRIGGE..

Result 15 Result 16 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
76	21:45:00	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned	0.000 sec / 0.000 sec
77	21:45:00	DROP TRIGGER CheckBalance	0 row(s) affected	0.015 sec

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17. PROCEDURE MANAGEMENT COMMANDS

```

371 • SHOW PROCEDURE STATUS
372 WHERE Db = 'BankDB';
373 • DROP PROCEDURE GetAllCustomers;
374
375
376
377
378
379
380

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation
----	------	------	---------	----------	---------	---------------	---------	----------------------	-----------

Result Grid

Form Editor

Field Types

Result 17

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
78	21:48:55	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	0 row(s) returned	0.015 sec / 0.000 sec
79	21:48:55	DROP PROCEDURE GetAllCustomers	0 row(s) affected	0.015 sec